

An Introduction to Linux System Administration

This series of articles, which explains various concepts and commands in Linux, is aimed at those new to Linux. We take a look at the basics of Linux to gain a better understanding of the power it has.



Linux is a major force in computing technology, powering everything from mobile phones and personal computers to supercomputers and servers. The job of a systems administrator is to manage the operations of a computer system. As most computing devices are powered by Linux, it makes sense to learn it. By the end of this article, you should be able to know and understand:

- Linux file systems
- File system hierarchy
- Linux online manual page
- Root/super user
- Handling files and directories

All the commands are demonstrated using a CentOS

Linux distro.

The Linux file system

A file system is a method of storing files on a hard disk. There are different types of file systems supported by Linux:

- Conventional disk file systems: ext2, ext3, ext4, XFS, Btrfs, JFS, NTFS, etc.

- Flash storage file systems: ubifs, JFFS2, YAFFS, etc.
- Special-purpose file systems: procfs, sysfs, tmpfs, debugfs, etc.

File system hierarchy standards

The Linux system stores files according to a standard layout called the 'file system hierarchy'. The most common Linux directory structure is shown in Figure 1.

Linux online manual page

One of the key features of Linux is that it provides online help about every single command. To access the Linux manual (man) page, type the following command:

```
[bhargab@localhost~]$man ls
```

This will provide the manual page of the `ls` command.

Root or super user

This is a special kind of user account, which holds all kinds of permissions to do any alteration to a program or service